

STEVEN WANG

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SKILLS

CAD & Simulations: CATIA V5, SOLIDWORKS, AutoCAD, STAR-CCM+, Ansys Mechanical, FEA, CFD, GD&T, tolerance stack-up
Software/Electrical: MATLAB/Simulink, Python, C++, Java, Circuit Analysis, EV Powertrains, HTML/CSS, Microsoft Office 365 Suite
Manufacturing: Manual lathe/mill, CNC, Composites, 3D-printing, Sheet Metal, Waterjet, Laser-Cutting, DFM/DFMA

EXPERIENCE

Mechanical Design Engineer

Schaeffler Motion Technologies AG & Co. KG

- Designed and modeled a hydraulic torque-test fixture in CATIA-V5 to evaluate the geartrain efficiency and resistance torque of a 230:1 ratio ball valve actuator gearbox, developing the complete mechanical setup from concept through 3D assembly.
- Engineered and functionally validated 6 CAD-modeled vibration test fixtures, producing fully dimensioned manufacturing drawings to secure active purge pumps at critical mounting angles during durability vibration testing.
- Resolved fixture, tolerance and mounting-angle issues through root-cause analysis and rapid design iteration in weekly partner reviews, cutting fixture rework time by 50% and eliminating 16+ hours of downtime for durability testing programs.
- Delivered a fabricated and operational hydraulic torque-test setup, benchmarking a 230:1 gearbox at 5 N·m and measuring 87% overall geartrain efficiency across four operation test runs.
- Engaged partners in weekly meetings to review design tolerances, failure analysis, and root-cause investigations, while managing project timelines and deadlines.

Aerodynamics Design Engineer and Driveline Team Member

UW Formula Electric FSAE

- Leading nosecone and front wing design meetings for the FSAE aerodynamics team, assigning tasks to members and coordinating with mechanical and electrical subteams on component placement to improve overall system efficiency.
- Authored an R&D paper comparing parallel and series cooling loop architecture trade offs, using hand calculations and published literature to find solutions prioritizing improved endurance run performance and greater weight savings.
- Designed and manufactured chassis enclosure mounts, Tractive System Status Indicator, and electrical enclosure jigs, applying poka-yoke to prevent misalignment while validating mounts using Ansys FEA meeting structural requirements.
- Manufactured carbon-fiber aerodynamic and structural components through 6+ vacuum-bagged wet-layup and resin-infusion cycles, completing mold preparation, ply layup, vacuum testing, infusion, curing, demolding, and finishing.

Fluid System and Cooling Loop Testing Engineer

Veritas/HDT Automotive Solutions AG

- Validated 500+ fuel, AC, and transmission lines for Ford, GM, and Stellantis, through thermal cycling, tensile, and burst testing using custom fixtures, detecting 6 defects among 4 major part failures among warranty parts.
- Revised LEAN prototypes for R&D, updating CAD models and running 6 new balljoint control arm setups utilizing Solidworks design and in-lab testing equipment to support quality assurance divisions.
- Tracked and analyzed 13 thermal chamber testing programs across 12 weeks, organizing useful tests and delivering part data information.

PROJECTS

FSAE Nosecone 2026 & 2027 | CFD, Star-CCM+

January 2026 - present

- Redesigned the nosecone assembly CAD to comply with 2027 regulations while optimizing front wing intake airflow and mitigating tire wake turbulence through custom whisker geometry, .
- Redirected front wing upwash to enhance side and rear wing downforce, successfully shifting the aerodynamic center of pressure rearward to increase high-speed cornering stability.
- Conducted extensive CFD simulations using **Star-CCM+** evaluating front wing configurations by systematically varying winglet span, angle of attack, chord lengths, trailing edge geometry, and camber profiles.

Drive Cycle Model | MATLAB, Simulink

June 2026 - present

- Developing a backward facing EV powertrain model in MATLAB/Simulink to simulate energy consumption over EPA drive cycles, integrating road load dynamics, a loss-derived motor efficiency map, and inverter losses.
- Connecting and researching relation between model loss outputs to prior R&D work on parallel versus series cooling architectures, looking for motor, inverter, and battery I²R loss relationship to cooling system sizing across duty cycles.

EDUCATION

University of Waterloo

Bachelor of Honors Mechanical Engineering